

BANG Meeting

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BANG Updates

11.07.2025

Bateman

Ba beam -> implanted on tape in detector zone -> observed coincidences -> fresh piece of tape cycled in after t_{cycle} to prevent contamination

$$\dot{N}_p = R - \lambda N_p$$

$$\dot{N}_d = \lambda_p N_p - \lambda_d N_d$$

$$N_p(0) = N_d(0) = 0$$

`scipy.integrate.solve_ivp`

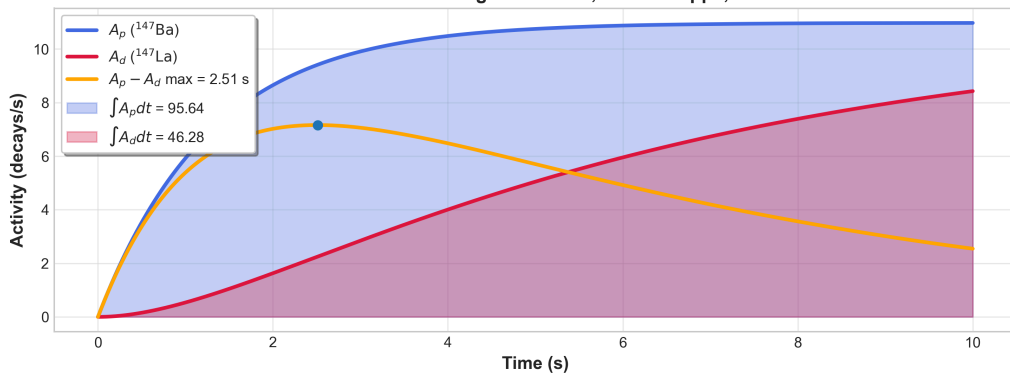
$$^{147}\text{Ba } t_{\frac{1}{2}} = 0.894 \text{ s} \quad ^{147}\text{La } t_{\frac{1}{2}} = 4.06 \text{ s}$$

$$^{148}\text{Ba } t_{\frac{1}{2}} = 0.619 \text{ s} \quad ^{148}\text{La } t_{\frac{1}{2}} = 1.411 \text{ s}$$

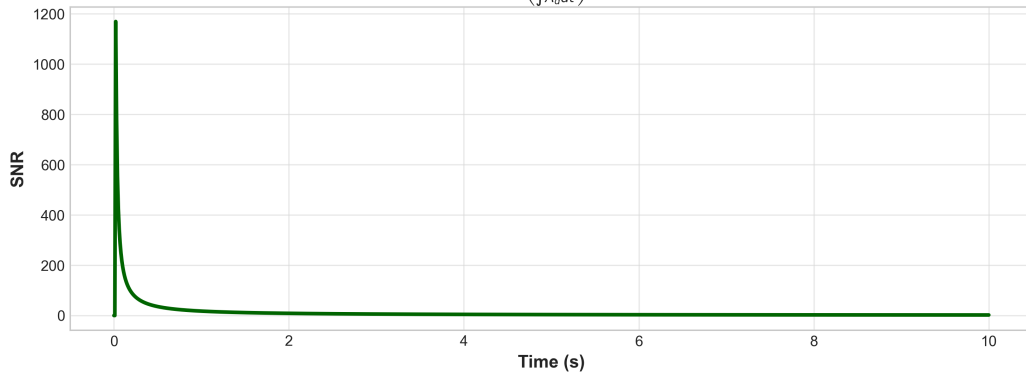
$$\text{maximize } \text{SNR}(t_{\text{cycle}}) = \frac{\int_0^{t_{\text{cycle}}} A_p dt}{\int_0^{t_{\text{cycle}}} A_d dt} \text{ might have to specify minimum contamination}$$

Find maximum net A_p

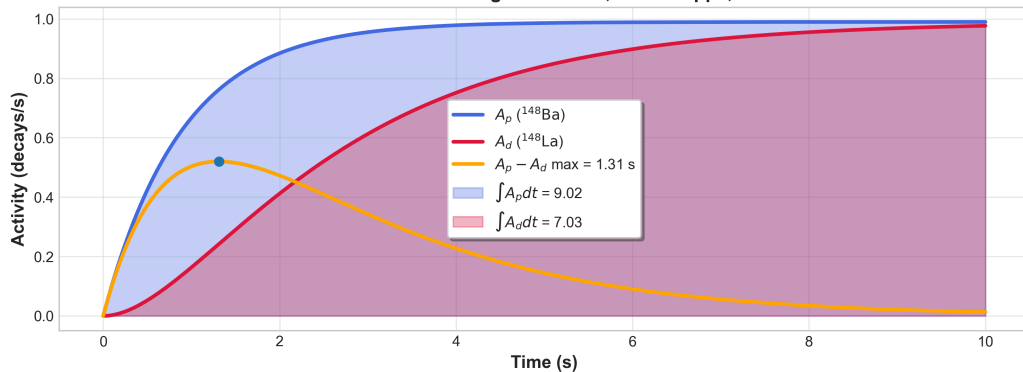
A = 147 Activities with Integrated Areas, R = 10.98 pps, SNR = 2.07



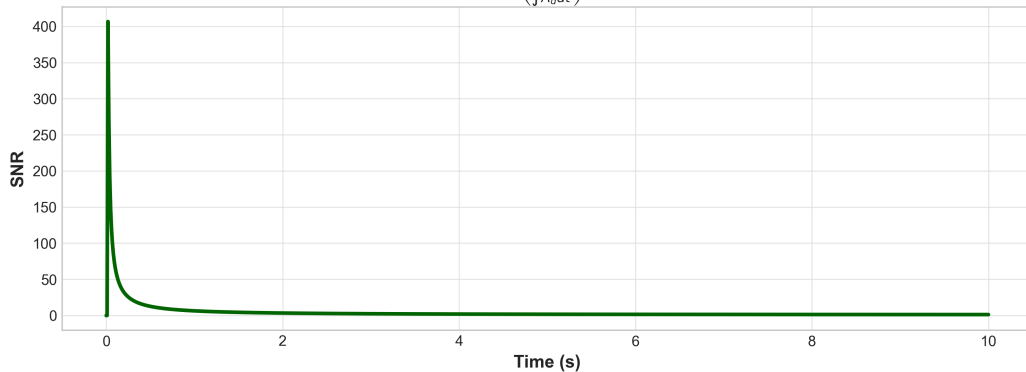
Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle



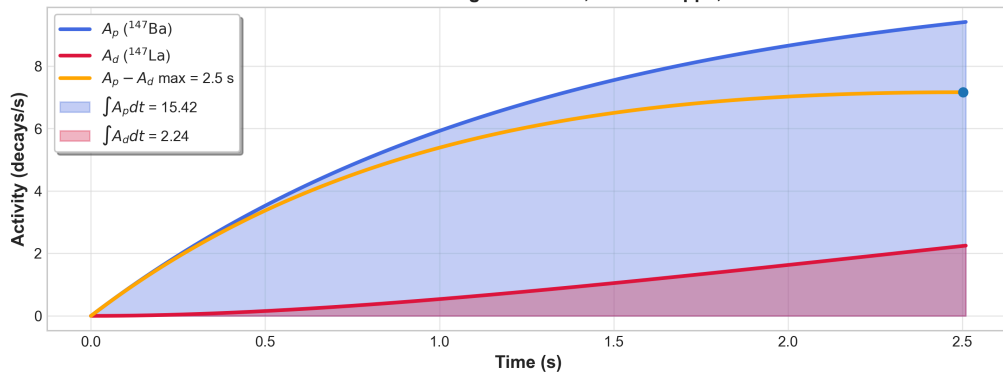
A = 148 Activities with Integrated Areas, R = 0.99 pps, SNR = 1.28



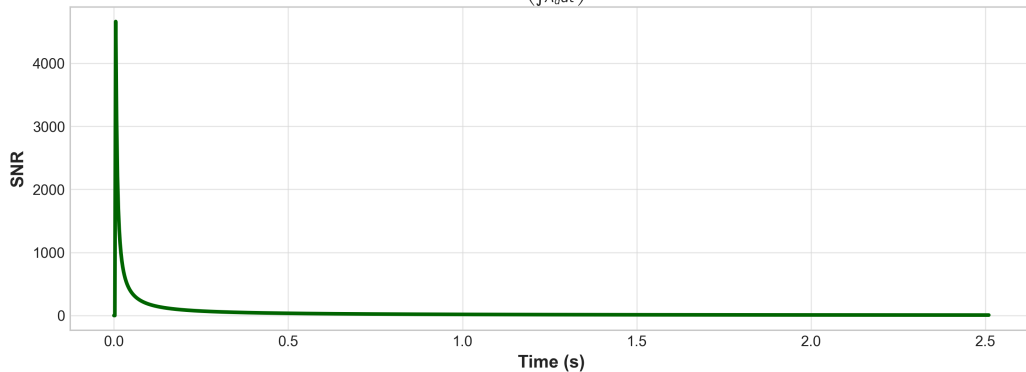
Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle



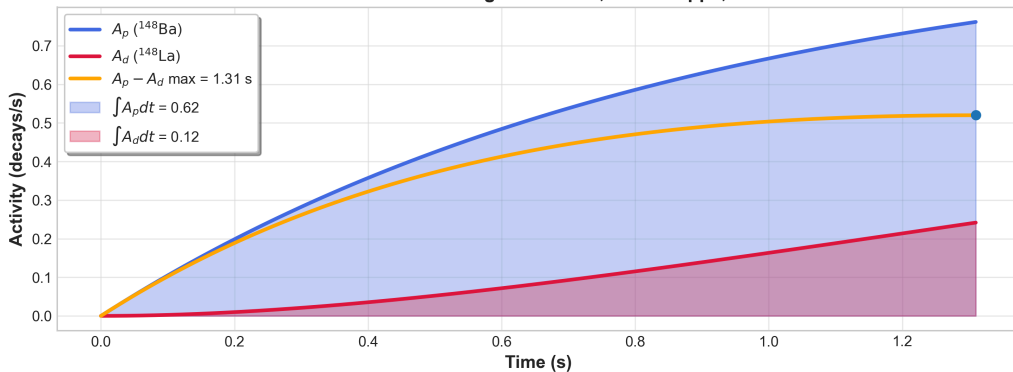
A = 147 Activities with Integrated Areas, R = 10.98 pps, SNR = 6.88



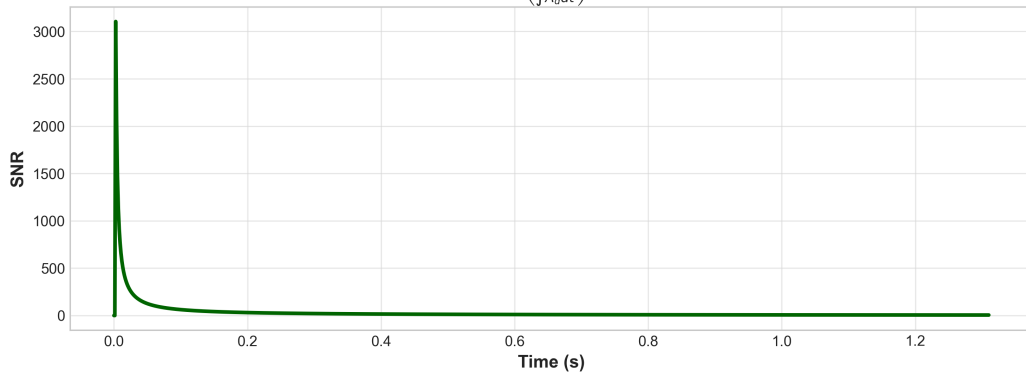
Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle



A = 148 Activities with Integrated Areas, R = 0.99 pps, SNR = 4.95



Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle



Notes

simulate detector response with monte carlo

- roll dice -> did it decay
 - did we observe it?
- optimize t_{cycle} over the entire experiment duration
 - observations will be periodic
- refer to adriannas literature for taping cycle times
- report total number of counts from the parent and daughter for various t_{cycle} s
 - then find the maximum

BANG Updates

11.14.2025

Tape timings

Previous experiments had daughter nuclei with minutes half-lives

$^{92}\text{Sr}(n, \gamma)^{93}\text{Sr}$ contamination level:

$$C = \frac{\int_0^{t_{\text{cycle}}} A_d(t) dt}{\int_0^{t_{\text{cycle}}} A_p(t) + A_d(t) dt} = 3.85\%$$

$$t_{\text{cycle}} = 60 \text{ s}$$

Found daughter $\beta - \gamma$ coincidence matrix and subtracted from the parent matrix

Will make monte carlo simulation